**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C5 · FLOATING AND FLYING

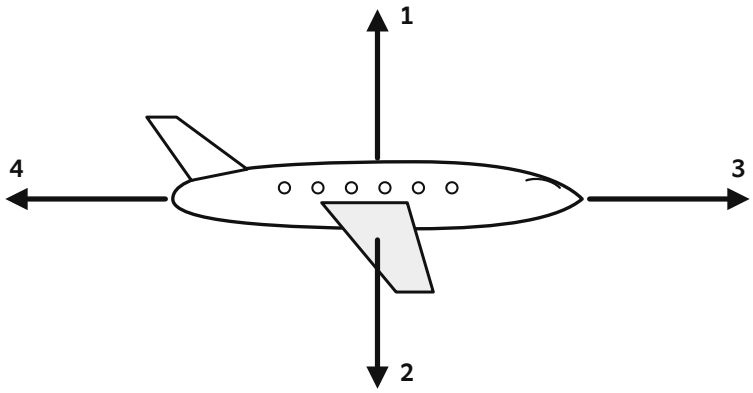
# Forces for floating and flying

★☆☆
Grade 4

**Learning goals**

- Label the forces of flight.
- Sort floaters and sinkers.

## 1 Label the four forces of flight. [4]




| 1 | 2 | 3 | 4 |
|---|---|---|---|
| lift | weight | thrust | drag |

## 2 Predict: float or sink? [8]

SORT THESE

**a glass marble** **a wooden stick** **a plastic duck** **a cork** **a coin** **a dry leaf** **a metal key** **a stone**

| Floats | Sinks |
|---|---|
| a cork | a stone |
| a plastic duck | a metal key |
| a dry leaf | a glass marble |
| a wooden stick | a coin |

## 3 Match the human invention to the living thing it copies. [4]

Draw a line to match each pair, or write the letter.

| | | |
|---|---|---|
| life jacket ● | → A | ● **A** duck feathers trap air |
| parachute ● | → C | ● **B** curved bird wing |
| aeroplane wing ● | → B | ● **C** fluffy seeds drifting on the wind |
| boat hull ● | → D | ● **D** floating coconut shell |

## 4 Fill in the gaps. [3]

WORD BANK

**gravity** **smaller** **bigger** **upthrust** **squashed**

The upward push of water is called upthrust. A stone sinks because its weight is bigger than the upthrust. Air can be squashed into a tyre.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C5 · FLOATING AND FLYING

# Why do things float and fly?

★★☆
Grade 4

**Learning goals**

- Explain floating with forces and density.
- Explain flight forces.

**1** **A steel nail sinks, but a steel ship floats. Explain why.** [3]

The ship is hollow and full of air, so it is light for its size and displaces a lot of water, giving enough upthrust to hold up its weight; the solid nail is too dense.

**2** **Why do life jackets help you float?** [2]

They are full of air or light foam, which makes you less dense overall, so the water can push you up enough to float.

**3** **A plane is taking off. Which forces must be bigger? Tick two.** [2]

| | True? |
|---|---|
| lift is bigger than weight | ✓ |
| weight is bigger than lift | |
| thrust is bigger than drag | ✓ |
| drag is bigger than thrust | |

**4** **Why does a parachute have such a big canopy?** [2]

A big canopy catches lots of air, making a big drag (air resistance) force, so the person falls slowly.

**5** **Draw a boat with arrows for weight and upthrust. Show what happens when the boat is overloaded.** [2]

Boat forces

Normal: weight and upthrust arrows equal. Overloaded: weight arrow longer; the boat sits lower and water may come in.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C5 · FLOATING AND FLYING

# Foil boats: which shape holds most?

★★☆
Grade 4

**Learning goals**

- Design and test model floaters.
- Record results.
- Improve a design.

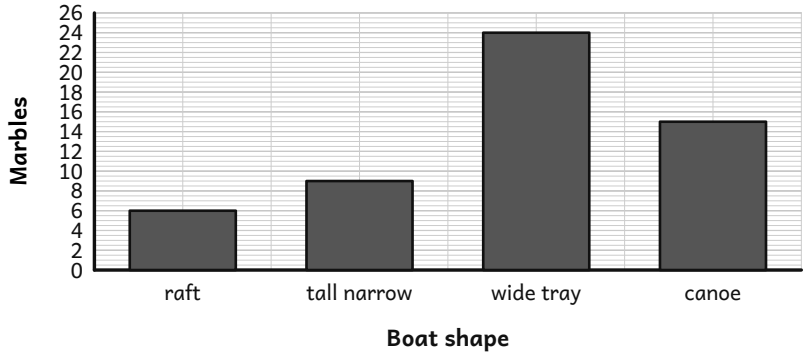
Class 4A made boats from the same size of aluminium foil (20 cm × 20 cm) in four shapes. They added marbles one at a time until each boat sank.

**1** **Complete the plan.** [3]

| | |
|---|---|
| Change | the shape of the boat |
| Measure | number of marbles before it sinks |
| Keep the same | size and type of foil, the marbles, the bowl of water, how marbles are added |

**2** **Draw a bar chart.** [3]

| Boat shape | flat raft | tall narrow | wide low tray | canoe |
|---|---|---|---|---|
| Marbles before sinking | 6 | 9 | 24 | 15 |




**3** **Which shape held the most? Explain using the word displace.** [2]

The wide low tray: it pushes aside (displaces) the most water, so the upthrust is biggest.

**4** **How could you improve the best boat?** [2]

Make the sides a little higher so water does not come in, or spread the marbles evenly so it stays balanced.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C5 · FLOATING AND FLYING

# Safe on the water: boats and launches

★★★
Grade 4

**Learning goals**

- Apply floating science to safety.
- Use data about loads.
- Suggest rules to save lives.

Science · Citizenship · Maths · Geography

Millions of people in Bangladesh travel by water every day: small *nouka*, ferries and big **launches** that carry hundreds of passengers between Dhaka and the south. When a boat is loaded, it sinks lower in the water. Every launch is allowed to carry only a certain number of people.

Sometimes, especially before holidays like Eid, launches are **overloaded**. The boat sits too low, waves can wash in, and it becomes unstable. In storms, overloaded boats can capsize. Safety rules say launches must not leave in bad weather, must carry life jackets and life buoys, and must not carry more people than allowed.

Based on river transport safety rules in Bangladesh. Data below are example values.

**1** **A launch is allowed to carry 300 passengers (example data).** [3]

| Trip | A | B | C | D |
|---|---|---|---|---|
| Passengers | 250 | 300 | 420 | 280 |

**a)** Which trip was overloaded? By how many people?

Trip C, by 120 people (420 – 300)

**b)** Why is an overloaded boat more likely to sink?

Its weight is bigger, so it sits lower; waves come over the side and fill it with water.

**2** **Many boats have a painted line on the side called a load line. What do you think it is for?** [2]

If the water reaches the line, the boat is carrying too much and must not leave.

**3** **What might each person say about overloading?** [3]

| Person | View |
|---|---|
| a passenger going home for Eid | I need to get home; there are not enough boats. |
| the launch owner | More passengers means more money, but it is dangerous and illegal. |
| a river police officer | I must stop overloaded boats to save lives. |

**4** **Design a safety poster for a launch terminal. Include two rules and a picture.** [2]

Safety poster

e.g. "Wear a life jacket", "Do not board a full launch", picture of a boat with the load line.

**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C5 · FLOATING AND FLYING

# Show what you know

★★★
Grade 4

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

## 1 Find the key words.

| S | T | H | R | U | S | T | A | I | R |
|---|---|---|---|---|---|---|---|---|---|
| B | L | W | L | F | L | O | A | T | D |
| Y | U | E | I | D | B | S | W | O | R |
| O | A | I | F | V | V | I | U | W | A |
| J | N | G | T | C | F | N | L | S | G |
| O | P | H | C | O | K | K | L | Y | B |
| S | L | T | K | B | D | Y | I | J | S |
| B | O | A | T | F | K | G | P | L | E |
| U | U | P | T | H | R | U | S | T | V |
| O | P | B | T | L | H | B | V | L | D |

**air boat drag float lift sink thrust upthrust weight**

## 2 Which force pulls an aeroplane backwards? [1]

- ○ lift
- ○ thrust
- ● **drag**
- ○ weight

## 3 Why does a block of wood float? [1]

- ○ it is heavy for its size
- ● **it is light for its size**
- ○ it has no weight
- ○ water pulls it up

## 4 Design a paper glider. Draw it and explain one change you could test to make it fly further. [3]

My glider

e.g. wider wings for more lift, or a paper clip on the nose to balance it; test by measuring the distance flown.

**5** **Challenge. A ball of clay sinks, but the same clay shaped into a bowl floats. Explain.** [2]

As a bowl it pushes aside much more water, so the upthrust is bigger and can match its weight. The mass is the same but the shape is different.